

Training and Testing

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can choose a split unit that matches the intended prediction claim.

I can keep related examples together and audit group overlap.

I can record split roles counts and limits before model evaluation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 11: Training and Testing

The test set estimates performance on unseen cases.



THINKER MISSION

Create a fair split before anyone sees the mystery cards.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Why does agreement between two annotators not prove their labels are correct?

This week's end goal

By the end, I can split related records by group when evaluation concerns new groups.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use training and testing to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A model may seem successful when its test examples closely resemble examples it already learned from. Imagine several pictures taken seconds apart during one robot trip. They have different picture IDs but share a route, lighting and objects. This week you will assign entire trips to separate sets and explain what the resulting evaluation could measure. No model is fitted in this lesson.

Our invented twelve-row dataset contains frame descriptions from six trips. Each row is one frame; the group is its trip ID. The proposed claim concerns new trips on the same practice route under similar conditions. We therefore keep all frames from a trip in one set. To claim success on new routes, grouping only by trip would not be enough.

Trip group · Frame IDs · Assigned role

Trip group: A

Frame IDs: A1 A2 A3

Assigned role: Training

Trip group: B

Frame IDs: B1 B2

Assigned role: Training

Trip group: C

Frame IDs: C1

Assigned role: Training

Trip group: D

Frame IDs: D1 D2

Assigned role: Validation

Trip group: E

Frame IDs: E1 E2

Assigned role: Test

Trip group: F

Frame IDs: F1 F2

Assigned role: Test

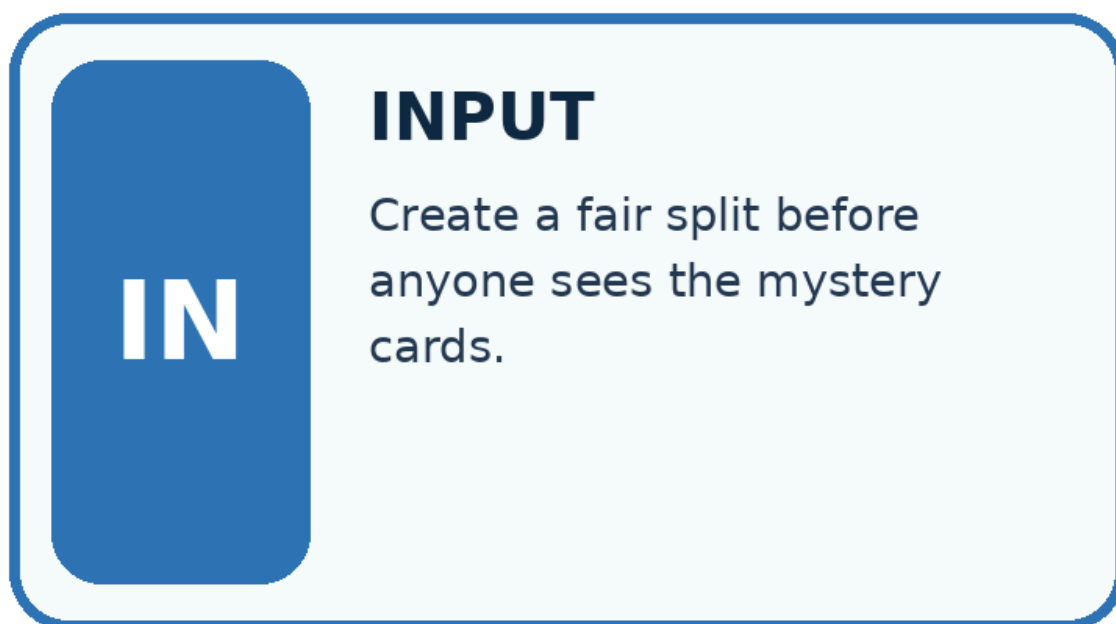
This small fictional assignment is for learning split design, not evidence of real performance. Each frame belongs to one group and one set. Store frame and trip IDs as metadata for auditing; do not assume an ID is an appropriate prediction input.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

The test set estimates performance on unseen cases.



OUT

OUTPUT

Create a fair split before anyone sees the mystery cards.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Training data fits the model. Validation data helps compare choices such as candidate rules. Keep the test set for a final check after those choices are fixed. A learner who repeatedly changes the model after seeing test results has used that set for development; its score is no longer an untouched final check.

```
train_groups = {"A", "B", "C"}
valid_groups = {"D"}
test_groups = {"E", "F"}

shared = ((train_groups & valid_groups)
          | (train_groups & test_groups)
          | (valid_groups & test_groups))

print(sorted(shared))

print(len(train_groups), len(valid_groups),
      len(test_groups))
```

The first line of output is []: no trip ID is shared between any pair of sets. The second is 3 1 2: training, validation and test contain three, one and two trip groups. The symbol & finds shared values and | combines sets. These are group counts, not frame counts.

Keep each trip in one role

Whole groups · Role · Purpose

Whole groups: A B C with all frames

Role: Training

Purpose: Fit from examples

Whole groups: D with all frames

Role: Validation

Purpose: Compare choices

Whole groups: E F with all frames

Role: Test

Purpose: Final check after choices

Keep related frames together. Separate group IDs prevent this form of overlap; they do not prove that every other source of leakage is absent.

This audit assumes the lists correctly describe every frame assignment. It cannot find an omitted frame, a wrong trip ID, a copied image with a different ID or a feature containing the answer. Check the assignment table and source records too.

Connect the picture and the explanation

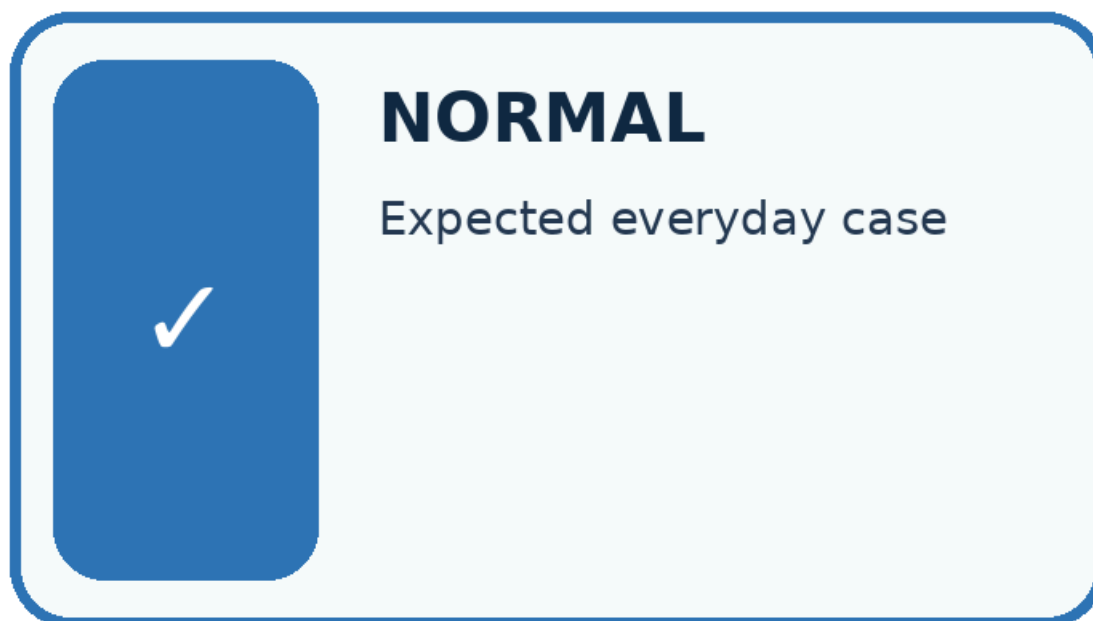
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





BOUNDARY

Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Give A2 a new filename without changing the book it depicts.

Prediction: Its group is still A; the original related examples still cross the split.

Reason: Renaming a row does not remove the relationship between scans.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

TRAINING SET

examples used to create or
adjust behavior

2

TEST SET

untouched examples used
for evaluation

3

UNSEEN

not used while designing

4

LEAKAGE

test information influences
the design

Word · My own example · Not this / boundary

Word: training set

My own example: Examples used to adjust a model's parameters or fit learned preprocessing.

Not this / boundary: _____

Word: test set

My own example: Data reserved for evaluating a selected system; final test data must not influence its

development.

Not this / boundary: _____

Word: unseen

My own example: Not previously used for the development, practice or selection being evaluated.

Not this / boundary: _____

Word: leakage

My own example: Use of information during development or prediction that should be unavailable for a fair evaluation of the intended task.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Validation set

Meaning: Separate examples used to compare choices during development

Word: Group

Meaning: Related examples that must stay together for a stated evaluation goal

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 11 • WORKED EVIDENCE

- 1 20 cards: seal 5 as test cases first.
- 2 Design the rule using only the remaining 15.
- 3 Freeze the rule before opening the test cards.
- 4 If a test label guided a change, it is no longer a fair final test.

1. 20 cards: seal 5 as test cases first.
2. Design the rule using only the remaining 15.
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4. If a test label guided a change, it is no longer a fair final test.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Suppose A1 and A2 are in training while A3 is in test. The frame IDs differ, but A appears in both sets. Shuffling the rows does not remove this relationship. For our new-trip claim, move all A frames to one chosen role and record the revised assignment before evaluating a model. Do not copy A3 into both roles.

Audit that every source frame occurs exactly once, no group crosses roles and the assigned counts sum to the source count. In our worked plan the row counts are six training, two validation and four test: twelve total. A fixed random seed can reproduce a random split, but it cannot make a flawed grouping choice fair.

A learner moves D2 from validation to training but leaves D1 in validation. Give the new row

counts and shared trip ID. Explain why different frame IDs and a fixed seed do not repair the overlap.

A split record names the prediction goal, source version, group definition, exact assignments, row and group counts, date and any random method or seed used. For this hand-assigned example, record that there was no random sampling. Keep label-rule versions and unresolved review statuses traceable as in Week 5.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Training and Testing

A fictional catalogue model learns from scans of book covers and will face previously unseen books.

The facts we will use

Book A has scans A1,A2; B has B1,B2; C has C1,C2; D has D1,D2.

Proposed training A1,B1,C1; validation A2,D1; test B2,C2,D2.

Training and Testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Choose unit

The intended kind of new example being evaluated is a new book, not merely a new scan.



2. Find leakage

A crosses train/validation; B,C cross train/test; D crosses validation/test.



3. Group repair

Training all A,B scans; validation all C scans; test all D scans.



4. Freeze assignment

Keep every scan from one book in exactly one role before fitting.

CHANGE ONE THING

Give A2 a new filename without changing the book it depicts.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Choose unit

What happens and why: The intended kind of new example being evaluated is a new book, not merely a new scan.

Step: 2. Find leakage

What happens and why: A crosses train/validation; B,C cross train/test; D crosses validation/test.

Step: 3. Group repair

What happens and why: Training all A,B scans; validation all C scans; test all D scans.

Step: 4. Freeze assignment

What happens and why: Keep every scan from one book in exactly one role before fitting.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Different scan IDs do not make examples independent books. A model may recognise repeated cover details from another view. Grouping by book matches the intended claim about new books and makes the split inspectable. The repaired split has four training, two validation and two test scans, but only two, one and one books respectively.

What this example does not establish

One validation book and one test book provide very limited diversity even though no groups overlap.

Read the labelled picture: How many distinct books are in each repaired role, and why report that as well as scan counts?

Change one thing: Give A2 a new filename without changing the book it depicts. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.

COMPARE • DIAGNOSE



Weak idea: Testing on training examples proves the system will generalize.
Strong move: locate evidence, change one cause, and retest.

Evidence record • Expected • Actual • Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Fit learned preprocessing, such as scaling values using an average, on training data and apply the stored transformation to the other sets. Using the full dataset to learn such values would let held out information influence development. We will implement preprocessing later; group separation alone does not solve that issue.

Record exclusions and label-review policies before scoring. Do not remove difficult test cases because they reduce the score. If a genuine data error requires repair, document it and its effect on the evaluation rather than claiming the old test stayed untouched. Next week you will build a simple baseline using these split roles.

All trip IDs are separate, but every trip uses the same route. Can you claim the model works on unseen routes? Explain the limitation and propose a split unit for that new claim.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

For new trips on a familiar route, holding out whole trips addresses within-trip similarity. For new routes, hold out entire routes so all their trips stay together. For forecasting future events, a time-respecting split may be needed: training on later events to predict earlier ones would not represent that intended use. The evaluation question determines the separation.

There is no universal percentage that makes a split valid. Large groups can make row counts unequal. Check that the sets can support the intended comparison, including enough relevant cases and labels, without moving individual related frames just to hit a percentage. This twelve-row example is too small to justify a reliable performance claim.

Practise with less help: Distinguish frames from groups

Use workbook W2 for the connected example “Splits and related examples”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

How many distinct books are in each repaired role, and why report that as well as scan counts?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Split cards before rule-making. Record the split, build on training only, freeze the rule, then reveal and score the test pile once.

Split cards before rule-making. Record the split, build on training only, freeze the rule, then reveal and score the test pile once.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Keep final user profiles hidden until weights and thresholds are fixed.

THIS WEEK'S CHECKPOINT Keep final user profiles hidden until weights and thresholds are fixed.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Group the recommender's related reviews, editions or repeated-user examples according to its intended evaluation claim.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Create a fair split before anyone sees the mystery cards.

CHALLENGE Create a fair split before anyone sees the mystery cards.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A rule is developed and evaluated using the same books A–H.

The question

Why does this not establish unseen performance?

Worked explanation

Those cases influenced the rule. Reserve different labelled books that do not influence construction or revision.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Separate development from fresh evaluation

Evidence or stage · Value or result

Evidence or stage: Books A–H

Value or result: Influenced the rule

Evidence or stage: Reserved books

Value or result: Did not influence construction

Evidence or stage: Revision

Value or result: Keep final cases separate

A rule is developed and evaluated using the same books A–H.

Why does this not establish unseen performance?

Those cases influenced the rule. Reserve different labelled books that do not influence construction or revision.

14. Try a new case independently

Independent case: Audit and repair independently

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is training P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared groups for each pair of sets. Repair it by assigning P and Q to training, R to validation and S to test. List all frames in each role and give row and group counts. State one limit of the repaired plan.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You tune a rule using cards 1 to 8 and test it on cards 9 and 10. After seeing card 9's error you change the rule. Is card 9 still untouched test evidence?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Keep final user profiles hidden until weights and thresholds are fixed.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A team tests five rule versions on the same ten books and chooses the highest score. It calls the final score an unbiased unseen test result. Explain the issue and revise the evaluation plan.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional catalogue model learns from scans of book covers and will face previously unseen books.

By the end, I can · How I will show it

By the end, I can: Choose a split unit that matches the intended prediction claim

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Keep related examples together and audit group overlap

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Record split roles counts and limits before model evaluation

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Choose a split unit that matches the intended prediction claim

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.